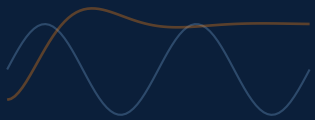
WEEK 1 · DAY 2 / 5

Mechatronics and Robotics Engineering (50 h)



Actuators II and Introduction to Control

Joint gearboxes ▪ Linear actuators & grippers ▪ Motor sizing ▪ Dynamic models ▪ Laplace & transfer functions

Mechatronics and Robotics Engineering Programme

Week 1 · Day 2 ▪ 2 weeks · 10 days · 50 hours

Agenda and learning outcomes

Code	Session	Time
1.5	Robot joint actuators: gearboxes, reflected inertia, backlash, stiffness, direct drive, SEA	1.0 h
1.6	Linear actuators and end-effectors	1.0 h
1.7	Motor sizing and selection for a robot joint	1.0 h
2.1	Modelling dynamic systems	1.0 h
2.2	Laplace transform and transfer functions	1.0 h
	Net teaching time	**5.0 h**

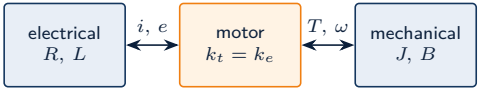
Breaks (≈10 min) roughly every hour are additional.

By the end of today you can . . .

- **explain** how planetary, harmonic and cycloidal gears work
- **calculate** reflected inertia, screw speed and gripper force
- **size** a motor for a robot joint (peak, RMS, speed)
- **model** mechanical and electromechanical systems as ODEs
- **find** a transfer function and read its poles

What we learned yesterday (W1D1)

- Motors turn **current into torque** (force on a wire) and **speed into voltage** (back-EMF).
- Robots mostly use electric actuators; pneumatics for simple grippers, hydraulics for very heavy loads.
- DC, BLDC and PMSM motors differ in how they **commutate** (brushes vs electronics).
- Steppers move in steps without feedback; induction motors drive conveyors and spindles.



$$V = Ri + L\frac{di}{dt} + k_e\omega$$

$$J\frac{d\omega}{dt} = k_t i - B\omega - T_L$$

Two time constants: electrical $\tau_e = L/R$ (ms) and mechanical $\tau_m = RJ/(k_t k_e)$ (tens of ms).

Today: what sits **between** the motor and the robot link – and how to describe the whole thing with equations.

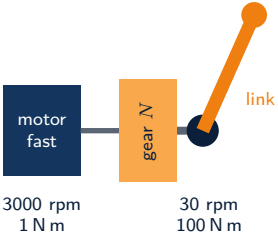
Session 1

1.5 Robot joint actuators: gearboxes, inertia, backlash, stiffness

Why a robot joint needs a gearbox



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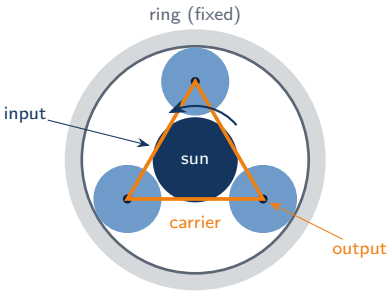


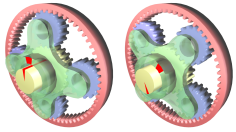
- Electric motors like **high speed, low torque**.
- Robot joints need **low speed, high torque**.
- A gearbox of ratio N trades one for the other – like the low gear on a bicycle.

$$N = \frac{\omega_m}{\omega_L}, \qquad T_L = \eta\, N\, T_m$$

η: efficiency (0.6–0.98). Typical joint: $N = 50$–160.

Planetary gearbox: sun, planets and ring





Wapcaplet, CC BY-SA 3.0, via Wikimedia Commons

- Input on the **sun**, ring held still, output on the **carrier** that holds the planets.
- Load shared by 3–4 planets: compact, stiff, efficient (95–98 %).

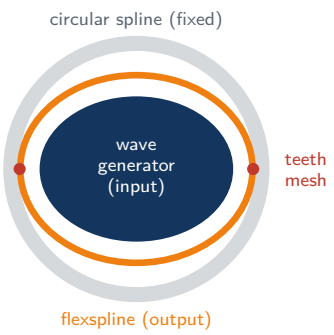
$$N = 1 + \frac{z_{\text{ring}}}{z_{\text{sun}}} \qquad (z\text{: number of teeth})$$

Worked example

1. $z_{\text{sun}} = 20$, $z_{\text{ring}} = 80$: $N = 1 + 4 = 5$.
2. Two stages in series: $N = 5 \cdot 5 = 25$.

Some play (backlash): 3–15 arcmin. Common on SCARA and cobot joints.

Harmonic drive (strain-wave gear): huge ratio, almost no play





Pieceofmetalwork, CC BY-SA 4.0, via Wikimedia Commons

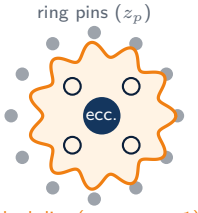
- An oval **wave generator** squeezes a thin toothed cup (**flexspline**) into a rigid ring with **2 more teeth**.
- Each input turn shifts the flexspline by 2 teeth.

$$N = \frac{z_{\text{flex}}}{z_{\text{circ}} - z_{\text{flex}}}$$

Example: $z_{\text{flex}} = 200$, $z_{\text{circ}} = 202 \Rightarrow N = 200/2 = 100$ in **one** stage.

Near-zero backlash, light, hollow shaft $\Rightarrow$ standard in arms and cobots. Drawbacks: springy, 70–85 % efficient.

Cycloidal drive – and how the three compare





1-1111 at German Wikipedia, Public domain, via Wikimedia Commons

An eccentric input makes a lobed disc "roll" inside a ring of pins; one lobe fewer than pins.

$$N = \frac{z_d}{z_p - z_d} \qquad \textbf{Example: } z_p = 40,\ z_d = 39 \Rightarrow N = 39$$

	Ratio / stage	Backlash	Stiffness & shock	Typical joint
Planetary	3–10	small	good	SCARA, wrist, AMR wheel
Harmonic	50–160	≈ 0	lower	cobot, small arm
Cycloidal	30–180	very small	high, shock-proof	base/shoulder of big arms

Going further: RV reducers combine a planetary first stage with a cycloidal second stage.

Reflected inertia: what the motor "feels"

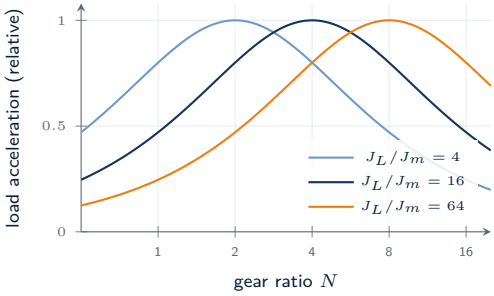
Kinetic energy is the same on both sides of the gear ($\omega_L = \omega_m / N$):

$$\tfrac{1}{2} J_L \omega_L^2 = \tfrac{1}{2} \frac{J_L}{N^2} \omega_m^2$$

$$J_{\text{tot}} = J_m + \frac{J_L}{N^2}, \qquad \text{load torque seen: } \frac{T_L}{N}$$

Worked example: robot link behind N 100

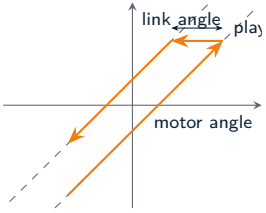
1. Link + payload: $J_L = 1.6\,\text{kg m}^2$.
2. Reflected: $1.6/100^2 = 1.6 \times 10^{-4}\,\text{kg m}^2$.
3. Motor $J_m = 0.18 \times 10^{-4} \Rightarrow$ **inertia ratio** $1.6/0.18 \approx 9$.



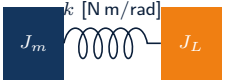
Fastest acceleration when $J_L/N^2 = J_m$, i.e. $N_{\text{opt}} = \sqrt{J_L/J_m}$.

Rule of thumb: keep the inertia ratio below ≈ 10 for easy tuning.

Backlash and torsional stiffness



Backlash: on reversal the motor turns but the link does not.



Every gearbox **twists** a little under load, like a stiff torsion spring:

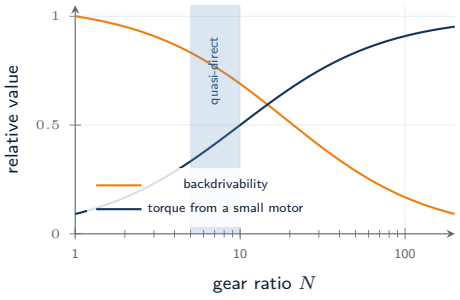
$$\Delta\theta = \frac{T}{k}$$

Worked example: harmonic drive

1. $k = 1 \times 10^4$ N m/rad, gravity torque $T = 34$ N m.
2. Twist: $34/10^4 = 3.4$ mrad.
3. At the tool, $r = 0.5$ m: $0.5 \cdot 3.4 = 1.7$ mm!
4. Backlash 1 arcmin $= 0.29$ mrad $\Rightarrow$ 0.15 mm.

Cures: preloaded gears, stiffer reducer, encoder on the link side (Day 9).

Direct drive and quasi-direct drive

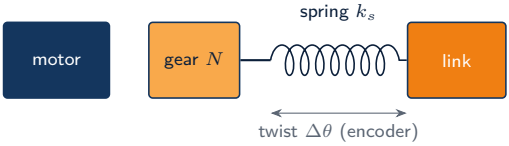


qualitative trend, not data

- **Direct drive** ($N = 1$): large torque motor bolted straight to the link – no backlash, no gear wear, very precise. But heavy and expensive.
- **Backdrivable**: pushing the link by hand turns the motor. Important for safe contact.
- **Quasi-direct drive** ($N \approx 6$–10): a compromise used in legged robots.
- High ratio ($N = 100$): small motor, but stiff to push and high friction.

Example: 34 N m with no gear needs a motor $\approx 100\times$ stronger than the 0.34 N m motor behind $N = 100$.

Series-elastic actuator (SEA): a spring on purpose



$$T = k_s \, \Delta\theta$$

Example: $k_s = 300\,\text{N m/rad}$, measured twist $0.05\,\text{rad}$ $\Rightarrow T = 15\,\text{N m}$.

Analogy: carrying a full cup with a soft wrist – bumps are absorbed, and you **feel** how hard you push.

- Measuring the spring twist = a cheap, robust **torque sensor**.
- Spring absorbs impacts $\Rightarrow$ safer for humans and gears.
- Price: lower **bandwidth** – the joint cannot react as fast.

Going further: variable-stiffness actuators; SEAs in exoskeletons and legged robots [Pratt & Williamson, 1995].

Session 1.5 – Check your understanding

Check your understanding

1. A planetary stage has $z_{\text{sun}} = 15$, $z_{\text{ring}} = 60$. What is its ratio?
2. A link with $J_L = 2\,\text{kg}\,\text{m}^2$ is driven through $N = 100$. What inertia does the motor feel?
3. Why do cobots often use harmonic drives, and why do some legged robots avoid high ratios?

Hint: $N = 1 + z_{\text{ring}}/z_{\text{sun}}$; inertia is divided by N^2; think about backlash and backdrivability.

Session 1.5 – Summary and key terms

- Gearboxes give the joint torque; motors give speed.
- Planetary (low ratio), harmonic (high ratio, no play), cycloidal (stiff, shock-proof).
- The load is reflected to the motor: inertia $\div N^2$, torque $\div N$.
- Backlash and finite stiffness cost accuracy at the tool.
- Direct drive and SEAs trade ratio for backdrivability and safety.

Remember

$$N = \omega_m/\omega_L, \qquad J_{\text{tot}} = J_m + J_L/N^2, \qquad \Delta\theta = T/k$$

Key terms

Gear ratio N: motor speed / joint speed.
Reflected inertia: load inertia felt at the motor.
Backlash: lost motion on reversal.
Torsional stiffness: torque per radian of twist.
Backdrivable: link can be pushed by hand.
SEA: actuator with a deliberate spring.

Session 2

1.6 Linear actuators and end-effectors

From rotation to straight-line motion

Mechanism	Speed	Stiffness	Accuracy	Robot use
Ball screw	$\leq$ 1–2 m/s	high	µm	Cartesian axes, SCARA Z
Toothed belt	$\leq$ 5–10 m/s	low	0.1 mm	long gantries
Rack & pinion	2–5 m/s	medium	0.05 mm	7th axis (track)
Linear motor	$\leq$ 5 m/s	highest	µm	wafer, PCB machines
Pneumatic cylinder	fast, end-to-end	low	two end points	simple pick & place

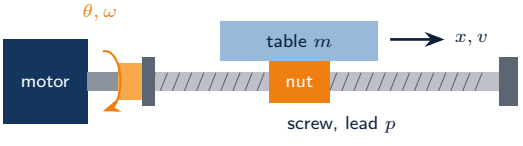
Rule of thumb

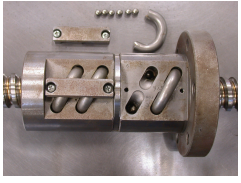
Rotary motor + screw/belt: cheap and flexible.
Linear motor: best performance, highest cost.

Every converter has a ratio

Screw: travel per turn (**lead** p). Belt and rack: pulley radius r.

Ball screw: a nut on a bolt, with rolling balls





Glenn McKechnie (Graibeard), CC BY-SA 3.0, via Wikimedia Commons

Turn the bolt once, the nut moves one **lead** p [m/rev]. Balls roll instead of slide: $\eta \approx 0.9$.

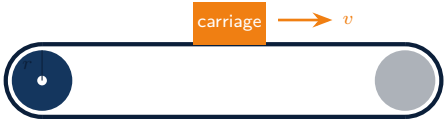
$$v = \frac{p}{2\pi}\,\omega, \qquad F = \frac{2\pi\,\eta}{p}\,T$$

Worked example: lead 10 mm

1. 3000 rpm $= 50$ rev/s $\Rightarrow$ $v = 50 \cdot 0.01 = 0.5$ m/s.
2. $T = 1$ N m: $F = 6.28 \cdot 0.9 / 0.01 \approx 565$ N.

A small motor pushes like a 57 kg weight!

Belt and rack drives: same idea, different ratio



$$v = r\,\omega, \qquad F = \frac{T}{r}, \qquad J_{\text{refl}} = m\,r^2$$

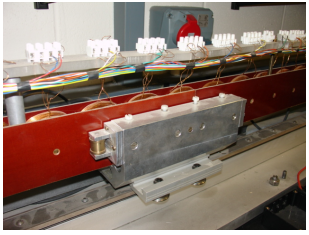
Rack & pinion: same formulas, r = pinion pitch radius.

Worked example: carry m 5 kg

1. Belt, pulley $r = 20\,\text{mm}$: $J_{\text{refl}} = 5 \cdot 0.02^2 = 2 \times 10^{-3}\,\text{kg m}^2$.
2. Ball screw, $p = 10\,\text{mm}$: $J_{\text{refl}} = m\,(p/2\pi)^2 = 5 \cdot (0.0016)^2 \approx 1.3 \times 10^{-5}\,\text{kg m}^2$.
3. Ratio of the two: ≈ 160!

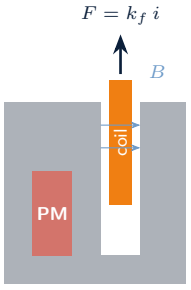
- A belt is a "**low gear ratio**": fast and long, but the motor feels much more of the load.
- That is why belt axes often add a small planetary gearbox.
- Belts stretch $\Rightarrow$ low stiffness, wobble at stops.

Linear motor and voice coil: no conversion needed



Zureks, CC BY-SA 4.0, via Wikimedia Commons

- **Linear motor** = a PMSM "unrolled" flat: coil carriage + magnet track.
- Direct drive for straight lines: stiff, fast, no play.



voice coil, half-section

Voice coil: like a loudspeaker – a coil in a magnet gap moves in and out.

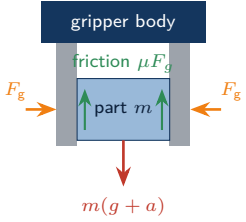
$$F = k_f\, i, \qquad m\frac{dv}{dt} = k_f\, i - b\, v - F_L$$

- k_f [N/A]: force constant.
- Same equations as the DC motor with $T \to F$, $\omega \to v$, $J \to m$.
- Example: $k_f = 20\,\text{N/A}$, $3\,\text{A} \Rightarrow 60\,\text{N}$.

Grippers: electric and pneumatic jaws



BPLINFO, CC BY-SA 4.0, via Wikimedia Commons



Two jaws squeeze; **friction** carries the weight (no form-fit).

$$2\,\mu\,F_g \;\geq\; S \cdot m\,(g+a)$$

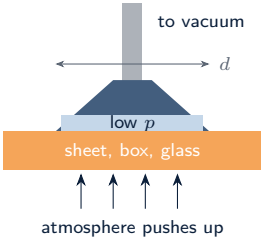
μ: friction coefficient; S: safety factor (2–4); a: vertical acceleration.

Worked example

1. $m = 2\,\text{kg}$, $a = 5\,\text{m/s}^2$: $m(g+a) = 2 \cdot 14.8 = 29.6\,\text{N}$.
2. With $S = 2$: 59 N must be held.
3. $\mu = 0.4$: $F_g \geq 59/(2 \cdot 0.4) \approx$ **74 N** per jaw.

Pneumatic: cheap, fast, open/closed only. **Electric**: programmable force and width, quieter, needs no air.

Vacuum grippers and suction cups



The cup does not "suck": the **air outside pushes** the part against it.

$$F = \Delta p \cdot A, \qquad A = \pi d^2/4$$

Worked example: lift a 5 kg box

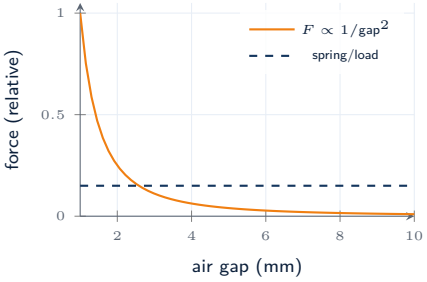
1. One cup, $d = 40$ mm: $A = 1.26 \times 10^{-3}$ m^2.
2. $\Delta p = 60$ kPa: $F = 60000 \cdot 0.00126 \approx 75$ N.
3. Needed: $5 \cdot (9.81 + 10) \cdot 2 \approx 198$ N ($a = 10$, $S = 2$).
4. Use **3 cups** (225 N).

Great for flat, smooth, light parts; porous or dusty surfaces leak.

Solenoids and pneumatic cylinders: simple two-position actuators



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A **solenoid** pulls hardest when almost closed – it "snaps" in.

Pneumatic cylinder: compressed air pushes a piston.

$$F = p \cdot A$$

1. Bore 32 mm: $A = 8.0 \times 10^{-4}\ \text{m}^2$.
2. $p = 6\ \text{bar} = 6 \times 10^5\ \text{Pa}$: $F \approx 480\ \text{N}$.

- Both are **bang-bang**: two end positions, no control in between.
- Uses: valves, locks, part stops, ejectors.

Session 1.6 – Check your understanding

Check your understanding

1. A ball screw has lead 5 mm. How fast does the nut move at 1200 rpm?
2. A suction cup of $1 \times 10^{-3}\,\mathrm{m}^2$ works at $\Delta p = 50\,\mathrm{kPa}$. What is its theoretical holding force?
3. Why does a belt axis "feel" heavier to the motor than a ball-screw axis with the same carriage?

Hint: $v = p \cdot n$; $F = \Delta p\, A$; compare mr^2 with $m(p/2\pi)^2$.

Session 1.6 – Summary and key terms

- Screws, belts and racks turn rotation into translation, each with its own ratio.
- Ball screw: precise and strong; belt: fast and long but soft.
- Linear motors and voice coils: direct drive for straight lines.
- Grippers hold by friction ($2\mu F_g$) or by air pressure ($\Delta p\, A$).
- Solenoids and cylinders: simple two-position motion.

Remember

$$v = \frac{p}{2\pi}\omega, \quad J = m\left(\frac{p}{2\pi}\right)^2, \quad J = mr^2, \quad F = \Delta p\, A$$

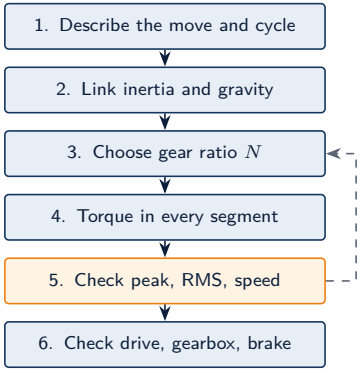
Key terms

Lead: travel per screw turn.
End-effector: tool at the robot flange.
Gripping force: squeeze force per jaw.
Safety factor: margin for unknowns.
Solenoid: coil that pulls an iron plunger.
Bang-bang: only two positions.

Session 3

1.7 Motor sizing and selection for a robot joint

What loads a robot joint motor?

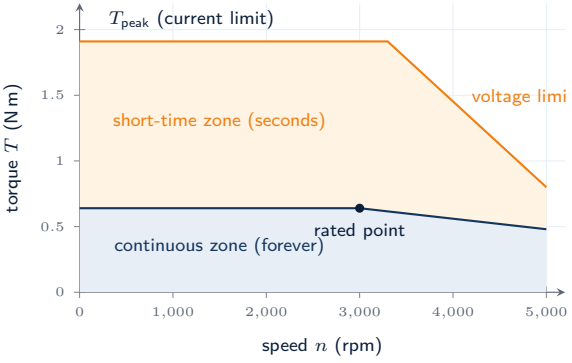


Motor torque = accelerate the inertia + hold the gravity + overcome friction:

$$T_m = \left(J_m + \frac{J_L}{N^2}\right)\alpha_m + \frac{T_g}{\eta N} + T_f$$

- $\alpha_m = N\alpha_L$: motor acceleration [rad/s^2].
- $T_g = mgr\cos\theta$: **gravity torque**, largest with the arm horizontal.
- In a robot, gravity often **dominates** – and acts even when standing still.
- Payload changes J_L and T_g – size for the worst case.

Servo motor torque–speed envelope

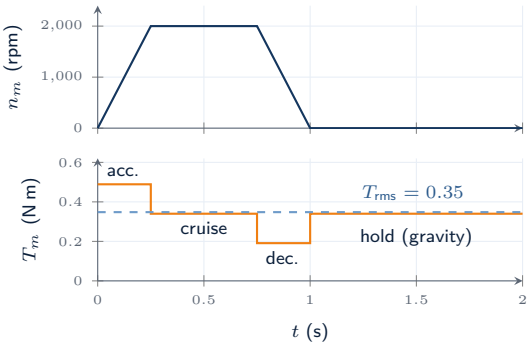


Typical 200 W servo: $T_N = 0.64$ N m, $T_{\text{peak}} = 1.91$ N m, $n_N = 3000$ rpm, $J_m = 0.18 \times 10^{-4}$ kg m^2.

Analogy: you can sprint for 10 seconds, but only jog for an hour.

- **Continuous torque**: possible forever without overheating.
- **Peak torque**: $\approx 3\times$ more, for short bursts (acceleration).
- **Voltage limit**: at high speed the back-EMF "eats" the supply voltage.

Duty cycle and RMS torque

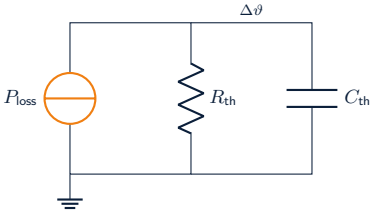


Heat $\propto i^2 \propto T^2$, and the motor heats up slowly, so it "feels" the **average of T^2**:

$$T_{\text{rms}} = \sqrt{\frac{1}{t_{\text{cyc}}} \sum_k T_k^2 \, t_k} \;\le\; T_{\text{cont}}$$

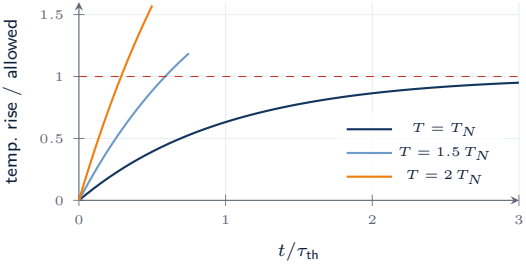
- T_k: torque in segment k, lasting t_k seconds.
- Why not the plain average? Negative torque heats just as much.
- Robot joint: holding against gravity also heats the motor.
- Also check $\max |T_k| \le T_{\text{peak}}$ and n_{max}.

Thermal limits: the motor warms like an RC circuit charges



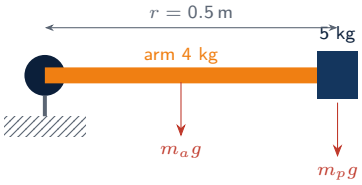
$$\Delta\vartheta(t) = P_{\text{loss}} R_{\text{th}} \left(1 - e^{-t/\tau_{\text{th}}}\right)$$

$\tau_{\text{th}} = R_{\text{th}} C_{\text{th}}$: minutes for a servo motor.



- R_{th} [K/W]: temperature rise per watt of loss; C_{th} [J/K]: heat capacity.
- Copper loss $\propto T^2$: twice the torque = **four times** the heat.
- Example: $P = 50\,\text{W}$, $R_{\text{th}} = 1.6\,\text{K/W} \Rightarrow$ final rise 80 K.

Worked example (1/2): shoulder-type joint, the load side



Given

Payload 5 kg at 0.5 m; arm 4 kg, uniform, 0.5 m.
Move 90°: accel 0.25 s, cruise 0.5 s, decel 0.25 s, hold 1.0 s.
Harmonic drive $N = 100$ (assume $\eta \approx 1$).

Step 1 – link inertia

$J_L = m_p r^2 + \frac{1}{3} m_a L^2 = 1.25 + 0.33 \approx 1.6\ \text{kg m}^2$

Step 2 – worst-case gravity torque (arm horizontal)

$T_g = 5 \cdot 9.81 \cdot 0.5 + 4 \cdot 9.81 \cdot 0.25 \approx 34\ \text{N m}$

Step 3 – joint speed and acceleration

$$\omega_L = \frac{\pi/2}{0.25 + 0.5} = 2.09\ \text{rad/s}, \qquad \alpha_L = \frac{2.09}{0.25} = 8.4\ \text{rad/s}^2$$

Step 4 – at the motor ($\times N$)

$\omega_m = 209\ \text{rad/s} = \mathbf{2000\ rpm}, \qquad \alpha_m = 838\ \text{rad/s}^2$

Worked example (2/2): motor torques and checks

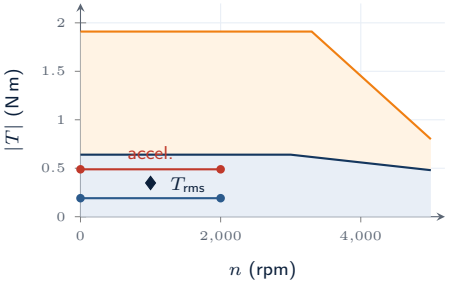
Step 5 – candidate: 200 W servo,
$J_m = 0.18 \times 10^{-4}$ kg m^2.
$J_{\text{tot}} = (0.18 + 1.60) \cdot 10^{-4} = 1.78 \times 10^{-4}$ kg m^2 (ratio ≈ 9)

Step 6 – torque pieces at the motor
inertia: $1.78\text{e-}4 \cdot 838 = 0.15$ N m; gravity: $34/100 = 0.34$ N m
acc. 0.49; cruise 0.34; dec. 0.19; hold 0.34 N m

Step 7 – RMS over the 2 s cycle

$$T_{\text{rms}} = \sqrt{\frac{0.49^2 \cdot 0.25 + 0.34^2 \cdot 0.5 + 0.19^2 \cdot 0.25 + 0.34^2 \cdot 1}{2}} = \mathbf{0.35\ \text{N m}}$$

Step 8 – checks
peak $0.49 < 1.91$ ✓ RMS $0.35 < 0.64$ ✓
$2000 \leq 3000$ rpm ✓



Gravity is 70 % of the RMS torque. A holding **brake** in the hold phase cuts T_{rms} to 0.25 N m. Also check the gearbox: output peak ≈ 49 N m.

Session 1.7 – Check your understanding

Check your understanding

1. In a 1 s cycle the motor gives 1 N m for 0.25 s and zero otherwise. What is T_{rms}?
2. In the worked example, the payload doubles to 10 kg. Which torque piece grows most, and is the 200 W motor still enough?
3. Why can a motor deliver $3\times$ its rated torque, but not permanently?

Hint: heat $\propto T^2$; gravity torque $\propto$ mass $\times$ distance.

Session 1.7 – Summary and key terms

- Joint torque = inertia + gravity + friction, all reflected through N.
- Peak torque must sit in the short-time zone, RMS torque in the continuous zone.
- The motor heats up like an RC circuit: exponentially, over minutes.
- In robots, holding against gravity can dominate the heating.
- Always check speed, drive current, gearbox rating and brake.

Remember

$$T_{\text{rms}} = \sqrt{\frac{1}{t_{\text{cyc}}} \sum T_k^2 t_k} \le T_{\text{cont}}, \qquad T_{\text{max}} \le T_{\text{peak}}$$

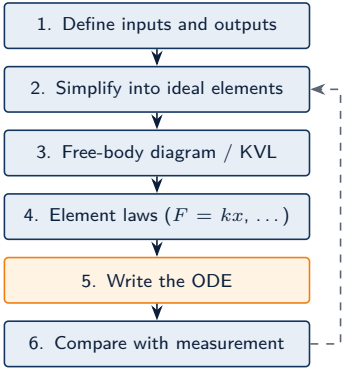
Key terms

Continuous torque: torque possible forever.
Peak torque: short-burst torque.
RMS torque: heating-equivalent torque.
Duty cycle: the repeating motion pattern.
Gravity torque: weight $\times$ lever arm.
Thermal resistance: K of rise per W.

Session 4

2.1 Modelling dynamic systems

Why model? From hardware to equations

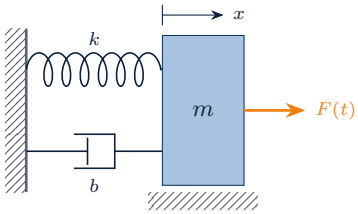


A controller can only be designed for something we can **predict**. A **model** predicts the output for a given input.

Domain	**Motion store**	**Position store**	**Loss**
Translation	mass m	spring k	damper b
Rotation	inertia J	torsion spring k_θ	damper B
Electrical	inductor L	capacitor C	resistor R

- **Lumped**: each property sits in one ideal element.
- Each energy store adds one to the **order** of the ODE.
- Use the **simplest model that answers your question**.

Mass–spring–damper: from free-body diagram to ODE



free-body diagram

Analogy: a car's suspension – spring carries the car, shock absorber (damper) stops the bouncing.

Newton $\sum F = m\ddot{x}$ with spring force kx and damper force $b\dot{x}$:

$$m\,\ddot{x} + b\,\dot{x} + k\,x = F(t)$$

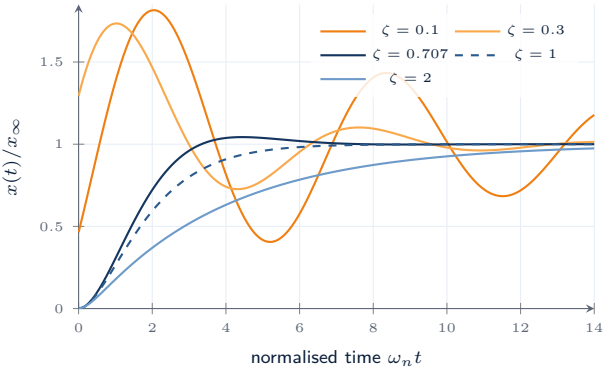
Two numbers describe it

$\omega_n = \sqrt{k/m}$ **natural frequency** [rad/s]: how fast it bounces

$\zeta = \dfrac{b}{2\sqrt{km}}$ **damping ratio** [–]: how fast bouncing dies

Example: $m = 2\,\text{kg}$, $k = 2 \times 10^5\,\text{N/m}$, $b = 120\,\text{N s/m}$:
$\omega_n = \sqrt{10^5} = 316\,\text{rad/s}$ (50 Hz),
$\zeta = 120/1265 = 0.095$.

Step responses: the role of damping



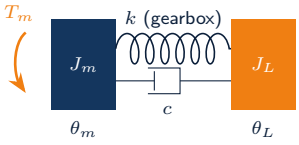
Push the mass with a constant force (a **step**) and watch $x(t)$:

- $\zeta < 1$: **overshoots and rings** – worn shock absorbers.
- $\zeta \approx 0.7$: fast, tiny overshoot (4 %) – a good target.
- $\zeta \geq 1$: no overshoot, but sluggish.

overshoot $\approx e^{-\pi\zeta/\sqrt{1-\zeta^2}}$

$\zeta = 0.1$: 73 %, $\zeta = 0.3$: 37 %.

Rotational systems: the flexible robot joint



$$J\,\ddot{\theta} + B\,\dot{\theta} + k_\theta\,\theta = T(t)$$

Same as mass–spring–damper with $m \to J$, $b \to B$, $k \to k_\theta$, $F \to T$.

- Newton for rotation: $\sum T = J\ddot{\theta}$.
- The gearbox is a torsion spring (Session 1.5) between motor and link.
- Holding the motor still, the link swings with $\omega_n = \sqrt{k/J_L}$.

Worked example: our shoulder joint

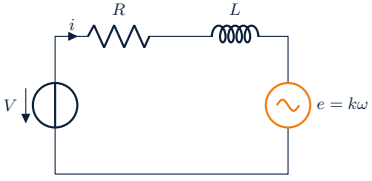
1. $k = 1 \times 10^4$ N m/rad, $J_L = 1.6$ kg m^2.
2. $\omega_n = \sqrt{10^4/1.6} = 79$ rad/s.
3. $f_n = 79/2\pi \approx$ **12.6 Hz** – low! The arm will wobble.

Going further: two-inertia resonance and notch filters (Session 3.7).

Electromechanical systems: one equation, three domains

Translation	Rotation	Electrical
force F	torque T	voltage v
velocity $\dot{x}$	speed ω	current i
mass m	inertia J	inductance L
damper b	damper B	resistance R
spring k	torsion k_θ	$1/C$

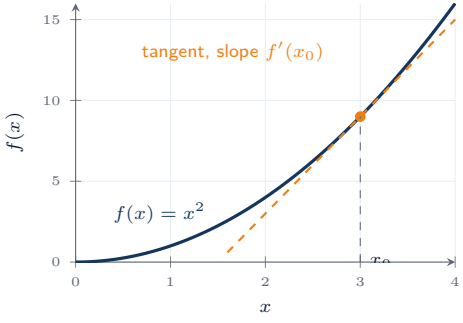
Always: (push) $\times$ (flow) $=$ **power**: Fv, $T\omega$, vi.



$$L\frac{di}{dt} + Ri = V - k\omega$$
$$J\frac{d\omega}{dt} + B\omega = k\,i - T_L$$

- Two coupled first-order ODEs $\Rightarrow$ 2nd-order system (stores L and J).
- The motor constant k is the bridge between the domains.

Linearisation: zoom in until the curve looks straight



Real robots are **nonlinear** (gravity $\propto \cos\theta$, friction, magnets). Close to an **operating point** x_0 every smooth curve looks like its tangent:

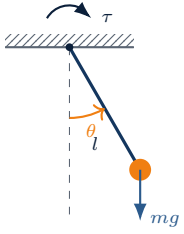
$$f(x) \approx f(x_0) + f'(x_0)\,(x - x_0)$$

Example $f = x^2$, $x_0 = 3$:

1. slope $f'(3) = 2 \cdot 3 = 6$
2. at $x = 3.1$: linear 9.6, exact 9.61 (0.1 % error)
3. at $x = 4$: linear 15, exact 16 (6 %) $\Rightarrow$ only **local**

Going further: many variables: Jacobian matrices $A = \partial \boldsymbol{f} / \partial \boldsymbol{x}$ (state space, Session 2.3).

Example: a robot link under gravity

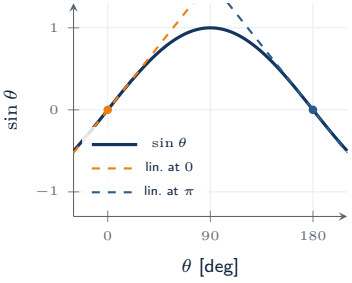


$$J\ddot{\theta} = \tau - mgl\sin\theta - b\dot{\theta}$$

1. Only $\sin\theta$ is nonlinear; its slope at θ_0 is $\cos\theta_0$.
2. Linear model for small deviations $\delta\theta$:

$$J\,\delta\ddot{\theta} + b\,\delta\dot{\theta} + (mgl\cos\theta_0)\,\delta\theta = \delta\tau$$

3. **Hanging** ($\cos 0 = +1$): gravity acts like a spring $\Rightarrow$ **stable**.
4. **Upright** ($\cos\pi = -1$): a "negative spring" $\Rightarrow$ **falls over**.



$\sin\theta \approx \theta$: 1 % error at 14°, 5 % at 31°.

Session 2.1 – Check your understanding

Check your understanding

1. You fit a gearbox that is $4\times$ stiffer. What happens to the natural frequency of the link?
2. A step response overshoots by 40 %. Should the damping b go up or down?
3. In the electrical/mechanical analogy, which element plays the role of the mass?

Hint: $\omega_n = \sqrt{k/m}$, $\zeta = b/(2\sqrt{km})$, and the analogy table.

Session 2.1 – Summary and key terms

- Lumped models: masses, springs, dampers (and L, C, R).
- Translation, rotation and circuits share one 2nd-order ODE with ω_n and ζ.
- A gearbox spring plus link inertia gives a low joint natural frequency.
- The motor couples the electrical and mechanical worlds.
- Nonlinear models are linearised around an operating point.

Remember

$$m\ddot{x} + b\dot{x} + kx = F, \quad \omega_n = \sqrt{k/m}, \quad \zeta = \frac{b}{2\sqrt{km}}$$

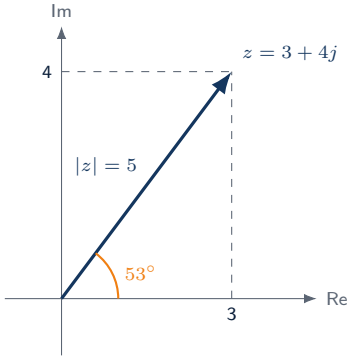
Key terms

Lumped model: built from ideal elements.
Natural frequency: speed of free oscillation.
Damping ratio: how fast oscillation dies.
Step response: reaction to a sudden input.
Operating point: where we linearise.

Session 5

2.2 Laplace transform and transfer functions

Refresher: complex numbers and the complex plane



Refresher: complex numbers

$z = a + jb$ (engineers write $j = \sqrt{-1}$). Polar form $z = |z|\, e^{j\varphi}$ with $|z| = \sqrt{a^2 + b^2}$, $\varphi = \operatorname{atan2}(b, a)$.

$$e^{j\varphi} = \cos\varphi + j\sin\varphi \qquad \text{(Euler)}$$

- Multiply: magnitudes multiply, angles add.
- Example: $\dfrac{1}{1+j}$: $|1+j| = 1.414$, $\angle 45° \Rightarrow 0.707\angle -45°$.
- For $s = \sigma + j\omega$: $e^{st} = e^{\sigma t}(\cos\omega t + j\sin\omega t)$ – σ sets growth/decay, ω sets oscillation.

Refresher: the Laplace transform as a tool

Refresher: Laplace transform

A "translation" $f(t) \to F(s)$ that turns **differential equations into algebra** – like logarithms turn multiplication into addition. We only use the table.

Time domain $f(t)$	Laplace $F(s)$
unit impulse $\delta(t)$	1
unit step $1(t)$	$1/s$
e^{-at}	$1/(s+a)$
derivative $\dot{f}$	$sF(s) - f(0)$
integral $\int_0^t f\, d\tau$	$F(s)/s$

Worked example: RC circuit, $RC = 1\,\text{ms}$, input u, capacitor voltage v (starting at 0 V):

1. ODE: $RC\,\dot{v} + v = u$.
2. Transform ($\dot{v} \to sV$): $RC\,s\,V + V = U$.
3. Solve for the ratio:

$$\frac{V(s)}{U(s)} = \frac{1}{RC\,s+1} = \frac{1}{0.001\,s+1}$$

4. 1 V step: $V = \frac{1}{s(0.001s+1)} \Rightarrow$ table: $v(t) = 1 - e^{-t/1\,\text{ms}}$ – the familiar RC charging curve.

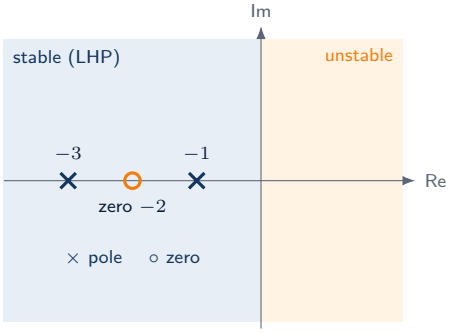
Transfer function, poles and zeros

The **transfer function** is the "recipe" of a system: output over input, starting from rest.

$$G(s) = \frac{Y(s)}{U(s)} = \frac{\text{numerator}(s)}{\text{denominator}(s)}$$

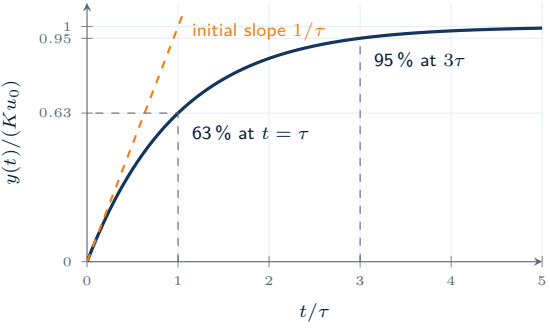
- **Poles**: roots of the denominator $\Rightarrow$ **stability and speed**.
- **Zeros**: roots of the numerator $\Rightarrow$ shape the response.
- **Order** = number of poles = number of energy stores.
- **DC gain** $G(0)$: output/input after everything settles.

Example: $G(s) = \dfrac{s+2}{(s+1)(s+3)}$: zero -2; poles $-1, -3$; $G(0) = 2/3$.



Left half-plane (LHP) = the stable side.

First-order system: one energy store

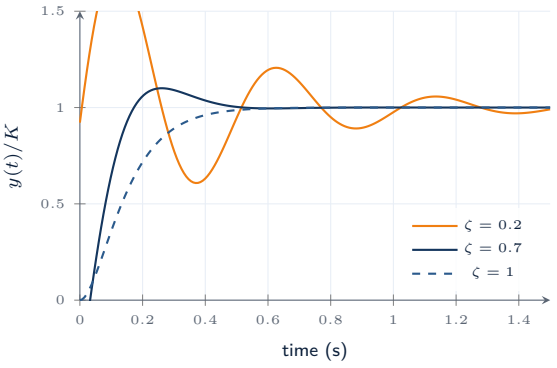


Like a cup of coffee cooling or an RC circuit charging.

$$G(s) = \frac{K}{\tau s + 1}, \quad y(t) = K u_0 \left(1 - e^{-t/\tau}\right)$$

- K: gain; τ: **time constant** [s]; pole at $s = -1/\tau$.
- 63 % after τ, 98 % after 4τ; never overshoots.
- Motor winding: $I/V = 1/(Ls + R)$, $\tau_e = L/R$ (2 mH/1 Ω = 2 ms).
- Flywheel with friction: $\Omega/T = 1/(Js + B)$, $\tau = J/B$.

Second-order system: the car suspension again



$\omega_n = 12.6\,\text{rad/s}$ (2 Hz) – computed responses.

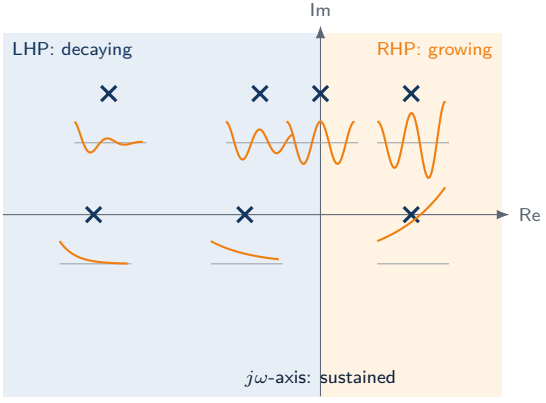
Mass m, spring k, damper b: $m\ddot{x} + b\dot{x} + kx = F$. Laplace: $(ms^2 + bs + k)X = F$.

$$G(s) = \frac{K\,\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- Two poles: $s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$.
- $\zeta < 1$: complex poles $\Rightarrow$ oscillation.
- $\zeta \geq 1$: two real poles $\Rightarrow$ no overshoot.

Quarter car: $m = 250\,\text{kg}$, $k = 16\,\text{kN/m}$, $b = 2000\,\text{N s/m} \Rightarrow \omega_n = 8$, $\zeta = 0.5$.

Where the poles are tells you how the system moves



A pole $p = -\sigma \pm j\omega_d$ gives a motion $e^{-\sigma t}\cos(\omega_d t)$:

- **Real part** $-\sigma$: how fast it dies out; settles in $\approx 4/\sigma$.
- **Imaginary part** ω_d: how fast it wiggles [rad/s].
- Right half-plane: grows $\Rightarrow$ **unstable**.
- Slowest poles (nearest the $j\omega$-axis) **dominate**.

Example: poles -2 and -20: the pole at -2 dominates; settling $\approx 4/2 = 2\,\text{s}$.

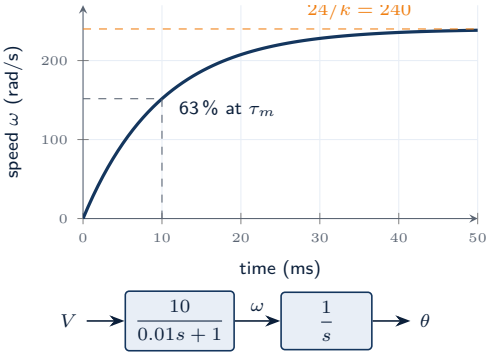
Worked example: transfer function of a robot joint motor

Neglect L (fast) and B in the DC-motor equations, then Laplace-transform:

$$\frac{\Omega(s)}{V(s)} = \frac{1/k}{\tau_m s + 1}, \qquad \tau_m = \frac{RJ}{k^2}$$

Worked example

1. $R = 1\,\Omega$, $k = 0.1\,\mathrm{N\,m/A}$, $J = 1 \times 10^{-4}\,\mathrm{kg\,m^2}$.
2. $\tau_m = 1 \cdot 10^{-4}/0.01 = 10\,\mathrm{ms}$; DC gain $1/k = 10\,\mathrm{rad/s/V}$.
3. 24 V step: final speed 240 rad/s, 63 % after 10 ms.
4. Position $\Theta = \Omega/s$ adds a pole at $s = 0$ (integrator).



Session 2.2 – Check your understanding

Check your understanding

1. Find the transfer function Y/U of $0.5\,\dot{y} + y = 2u$. What are its gain, time constant and pole?
2. A system has poles at $-1 \pm 5j$. Does it oscillate? Is it stable? Roughly how long until it settles?
3. Why does the joint *position* (not speed) keep growing after a constant voltage step?

Hint: $\dot{y} \to sY$; settling $\approx 4/\sigma$; integrating a constant gives a ramp.

Session 2.2 – Summary and key terms

- Laplace turns linear ODEs into algebra: $d/dt \to s$.
- Transfer function $G(s) = Y(s)/U(s)$; poles decide stability and speed.
- First order: $K/(\tau s + 1)$, 63 % at τ, no overshoot.
- Second order: ω_n and ζ; complex poles oscillate.
- Motor speed is first order; joint position adds an integrator.

Remember

$$\frac{K}{\tau s + 1}, \qquad \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \qquad \text{LHP poles} \Rightarrow \text{stable}$$

Key terms

Laplace transform: ODE $\to$ algebra.
Transfer function: output/input in s.
Pole / zero: root of denominator / numerator.
Time constant: time to reach 63 %.
DC gain: steady-state output/input.
Integrator: pole at $s = 0$.

Day summary

1.5 Gearboxes (planetary, harmonic, cycloidal) give joint torque; inertia is reflected by $1/N^2$; backlash and stiffness limit accuracy.

1.6 Screws, belts and linear motors make straight lines; grippers hold by friction or vacuum.

1.7 Size a joint motor with peak torque, RMS torque and speed – gravity matters.

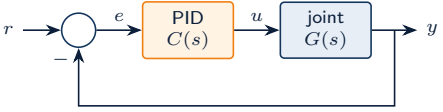
2.1 Mechanical, rotational and electrical systems share the same ODEs (ω_n, ζ).

2.2 Laplace and transfer functions: poles tell us speed, oscillation and stability.

The thread of the day

Motor + gear + link $\rightarrow$ equations $\rightarrow$ transfer function. Tomorrow we **close the loop** around it.

Preview of W1D3: Control in robotic and mechatronic systems



- 2.3 Block diagrams, feedback and state space
- 2.4 Frequency response and Bode plots
- 2.5 PID control: P, I, D, tuning, anti-windup, feedforward
- 2.6 Stability and performance: margins, bandwidth, steady-state error

Prepare: revise today's poles and ω_n, ζ.

Figure credits

Slide	File (Wikimedia Commons)	Author	License
Why a joint needs a gearbox	KUKA industrial robotic arm … 02.jpg	Oleg Yunakov	CC BY-SA 4.0
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All other drawings, plots and diagrams: own work for this course (TikZ / pgfplots / circuitikz).

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